

Supplementary Material

A Family of Classifying Criteria Emerging from Least-Squares Matching Between Partition Incidence Matrices

Boris Mirkin, Azizbek Bekniyazov

Results of Computational Experiments

Synthetic data

Synthetic datasets used in the experiments.

Dataset	V	L	α
1	6	4	0.5
2	6	4	0.85
3	15	4	0.5
4	15	4	0.85
5	6	8	0.5
6	6	8	0.85
7	15	8	0.5
8	15	8	0.85
9	6	15	0.5
10	6	15	0.85
11	15	15	0.5
12	15	15	0.85

Table 1.1: $N = 2000$, $V = 6$, $L = 4$, $\alpha = 0.5$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9915	0.9933	0.9933	0.9936	0.9934	0.9929	0.9935	0.9934	0.9936
	Std	0.0038	0.0030	0.0030	0.0031	0.0030	0.0033	0.0028	0.0029	0.0031
Precision	Mean	0.9916	0.9934	0.9934	0.9937	0.9935	0.9930	0.9935	0.9935	0.9936
	Std	0.0038	0.0029	0.0029	0.0031	0.0030	0.0032	0.0028	0.0028	0.0031
Recall	Mean	0.9915	0.9933	0.9933	0.9936	0.9934	0.9929	0.9935	0.9934	0.9936
	Std	0.0038	0.0030	0.0030	0.0031	0.0030	0.0033	0.0028	0.0029	0.0031
F1 score	Mean	0.9915	0.9933	0.9933	0.9936	0.9934	0.9929	0.9935	0.9934	0.9936
	Std	0.0038	0.0030	0.0030	0.0031	0.0030	0.0033	0.0028	0.0029	0.0031
ARI	Mean	0.9775	0.9822	0.9822	0.9831	0.9824	0.9812	0.9827	0.9824	0.9830
	Std	0.0099	0.0077	0.0077	0.0081	0.0079	0.0086	0.0072	0.0076	0.0080

Table 1.2: $N = 2000$, $V = 6$, $L = 4$, $\alpha = 0.5$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9926	0.9932	0.9931	0.9933	0.9935	0.9935	0.9931	0.9933	0.9932
	Std	0.0040	0.0038	0.0037	0.0037	0.0033	0.0035	0.0034	0.0036	0.0037
Precision	Mean	0.9927	0.9933	0.9932	0.9934	0.9936	0.9935	0.9931	0.9933	0.9932
	Std	0.0040	0.0038	0.0036	0.0036	0.0033	0.0034	0.0033	0.0036	0.0037
Recall	Mean	0.9926	0.9932	0.9931	0.9933	0.9935	0.9935	0.9931	0.9933	0.9932
	Std	0.0040	0.0038	0.0037	0.0037	0.0033	0.0035	0.0034	0.0036	0.0037
F1 score	Mean	0.9926	0.9932	0.9931	0.9933	0.9935	0.9935	0.9931	0.9933	0.9932
	Std	0.0040	0.0038	0.0037	0.0037	0.0033	0.0035	0.0034	0.0036	0.0037
ARI	Mean	0.9804	0.9821	0.9817	0.9823	0.9828	0.9827	0.9817	0.9821	0.9819
	Std	0.0104	0.0100	0.0095	0.0095	0.0087	0.0090	0.0086	0.0094	0.0096

Table 2.1: $N = 2000$, $V = 6$, $L = 4$, $\alpha = 0.85$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.7194	0.7312	0.7254	0.7356	0.7245	0.7245	0.7186	0.7271	0.7334
	Std	0.0216	0.0196	0.0218	0.0182	0.0212	0.0183	0.0250	0.0223	0.0183
Precision	Mean	0.7432	0.7408	0.7400	0.7440	0.7381	0.7388	0.7146	0.7386	0.7401
	Std	0.0188	0.0184	0.0188	0.0176	0.0203	0.0207	0.0342	0.0191	0.0178
Recall	Mean	0.7194	0.7312	0.7254	0.7356	0.7245	0.7245	0.7186	0.7271	0.7334
	Std	0.0216	0.0196	0.0218	0.0182	0.0212	0.0183	0.0250	0.0223	0.0183
F1 score	Mean	0.7205	0.7300	0.7259	0.7337	0.7246	0.7248	0.7048	0.7264	0.7300
	Std	0.0215	0.0191	0.0198	0.0179	0.0200	0.0182	0.0353	0.0206	0.0185
ARI	Mean	0.4245	0.4544	0.4393	0.4642	0.4383	0.4362	0.4635	0.4459	0.4629
	Std	0.0383	0.0377	0.0427	0.0332	0.0423	0.0376	0.0308	0.0436	0.0341

Table 2.2: $N = 2000$, $V = 6$, $L = 4$, $\alpha = 0.85$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.7451	0.7576	0.7529	0.7567	0.7518	0.7493	0.7446	0.7517	0.7529
	Std	0.0166	0.0160	0.0173	0.0161	0.0185	0.0187	0.0171	0.0161	0.0160
Precision	Mean	0.7567	0.7628	0.7616	0.7608	0.7627	0.7617	0.7565	0.7565	0.7566
	Std	0.0187	0.0175	0.0196	0.0175	0.0207	0.0204	0.0201	0.0181	0.0170
Recall	Mean	0.7451	0.7576	0.7529	0.7567	0.7518	0.7493	0.7446	0.7517	0.7529
	Std	0.0166	0.0160	0.0173	0.0161	0.0185	0.0187	0.0171	0.0161	0.0160
F1 score	Mean	0.7429	0.7571	0.7533	0.7553	0.7529	0.7508	0.7448	0.7503	0.7507
	Std	0.0181	0.0164	0.0173	0.0163	0.0180	0.0183	0.0173	0.0172	0.0175
ARI	Mean	0.4753	0.4937	0.4810	0.4960	0.4769	0.4707	0.4798	0.4838	0.4894
	Std	0.0331	0.0273	0.0316	0.0286	0.0337	0.0356	0.0300	0.0287	0.0268

Table 3.1: $N = 2000$, $V = 15$, $L = 4$, $\alpha = 0.5$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9994	0.9977	0.9975	0.9978	0.9974	0.9975	0.9982	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0022	0.0026	0.0024
Precision	Mean	0.9994	0.9977	0.9975	0.9978	0.9975	0.9975	0.9983	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0022	0.0025	0.0024
Recall	Mean	0.9994	0.9977	0.9975	0.9978	0.9974	0.9975	0.9982	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0022	0.0026	0.0024
F1 score	Mean	0.9994	0.9977	0.9975	0.9978	0.9974	0.9975	0.9982	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0022	0.0026	0.0024
ARI	Mean	0.9985	0.9939	0.9934	0.9941	0.9933	0.9935	0.9954	0.9930	0.9941
	Std	0.0027	0.0065	0.0070	0.0064	0.0066	0.0064	0.0059	0.0068	0.0063

Table 3.2: $N = 2000$, $V = 15$, $L = 4$, $\alpha = 0.5$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9994	0.9977	0.9975	0.9978	0.9975	0.9976	0.9984	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0023	0.0026	0.0024
Precision	Mean	0.9994	0.9977	0.9975	0.9978	0.9975	0.9976	0.9984	0.9974	0.9978
	Std	0.0010	0.0025	0.0025	0.0024	0.0024	0.0023	0.0022	0.0025	0.0024
Recall	Mean	0.9994	0.9977	0.9975	0.9978	0.9975	0.9976	0.9984	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0023	0.0026	0.0024
F1 score	Mean	0.9994	0.9977	0.9975	0.9978	0.9975	0.9976	0.9984	0.9974	0.9978
	Std	0.0010	0.0025	0.0026	0.0024	0.0025	0.0024	0.0023	0.0026	0.0024
ARI	Mean	0.9985	0.9939	0.9935	0.9941	0.9935	0.9937	0.9957	0.9930	0.9941
	Std	0.0027	0.0065	0.0068	0.0064	0.0065	0.0063	0.0059	0.0068	0.0063

Table 4.1: $N = 2000$, $V = 15$, $L = 4$, $\alpha = 0.85$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9515	0.9514	0.9502	0.9521	0.9490	0.9464	0.9428	0.9498	0.9509
	Std	0.0068	0.0076	0.0088	0.0079	0.0102	0.0147	0.0311	0.0085	0.0082
Precision	Mean	0.9527	0.9525	0.9514	0.9532	0.9502	0.9477	0.9466	0.9512	0.9521
	Std	0.0065	0.0074	0.0085	0.0076	0.0101	0.0143	0.0167	0.0080	0.0077
Recall	Mean	0.9515	0.9514	0.9502	0.9521	0.9490	0.9464	0.9428	0.9498	0.9509
	Std	0.0068	0.0076	0.0088	0.0079	0.0102	0.0147	0.0311	0.0085	0.0082
F1 score	Mean	0.9515	0.9515	0.9502	0.9522	0.9490	0.9464	0.9416	0.9499	0.9510
	Std	0.0068	0.0076	0.0088	0.0079	0.0102	0.0147	0.0405	0.0084	0.0081
ARI	Mean	0.8746	0.8744	0.8715	0.8763	0.8686	0.8621	0.8582	0.8700	0.8726
	Std	0.0168	0.0189	0.0219	0.0196	0.0256	0.0367	0.0417	0.0212	0.0205

Table 4.2: $N = 2000$, $V = 15$, $L = 4$, $\alpha = 0.85$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9660	0.9640	0.9627	0.9655	0.9627	0.9618	0.9572	0.9608	0.9608
	Std	0.0078	0.0084	0.0087	0.0089	0.0089	0.0092	0.0087	0.0083	0.0085
Precision	Mean	0.9666	0.9646	0.9634	0.9661	0.9633	0.9624	0.9582	0.9616	0.9615
	Std	0.0076	0.0082	0.0085	0.0087	0.0088	0.0091	0.0084	0.0083	0.0080
Recall	Mean	0.9660	0.9640	0.9627	0.9655	0.9627	0.9618	0.9572	0.9608	0.9608
	Std	0.0078	0.0084	0.0087	0.0089	0.0089	0.0092	0.0087	0.0083	0.0085
F1 score	Mean	0.9660	0.9640	0.9628	0.9656	0.9627	0.9618	0.9573	0.9609	0.9608
	Std	0.0078	0.0083	0.0087	0.0089	0.0089	0.0092	0.0087	0.0083	0.0084
ARI	Mean	0.9113	0.9061	0.9029	0.9101	0.9029	0.9006	0.8892	0.8980	0.8977
	Std	0.0199	0.0212	0.0221	0.0225	0.0229	0.0236	0.0219	0.0210	0.0218

Table 5.1: $N = 2000$, $V = 6$, $L = 8$, $\alpha = 0.5$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.8639	0.6444	0.6156	0.8579	0.6025	0.5420	0.7398	0.6181	0.6470
	Std	0.0158	0.0730	0.0435	0.0334	0.0309	0.0616	0.0428	0.0435	0.0757
Precision	Mean	0.8033	0.5973	0.5711	0.7949	0.5722	0.5622	0.6743	0.5868	0.6136
	Std	0.0190	0.1088	0.0917	0.0477	0.0833	0.0843	0.0644	0.1007	0.1117
Recall	Mean	0.8639	0.6444	0.6156	0.8579	0.6025	0.5420	0.7398	0.6181	0.6470
	Std	0.0158	0.0730	0.0435	0.0334	0.0309	0.0616	0.0428	0.0435	0.0757
F1 score	Mean	0.8236	0.5575	0.5375	0.8160	0.5317	0.4821	0.6775	0.5362	0.5600
	Std	0.0188	0.0866	0.0490	0.0434	0.0328	0.0596	0.0491	0.0495	0.0896
ARI	Mean	0.8466	0.6129	0.5236	0.8406	0.4729	0.3596	0.6552	0.5399	0.6124
	Std	0.0181	0.0944	0.0951	0.0316	0.0717	0.0916	0.0639	0.0910	0.0924

Table 5.2: $N = 2000$, $V = 6$, $L = 8$, $\alpha = 0.5$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9896	0.8782	0.8020	0.9888	0.7602	0.6780	0.8769	0.8166	0.8800
	Std	0.0038	0.0676	0.0723	0.0049	0.0556	0.0634	0.0423	0.0703	0.0699
Precision	Mean	0.9899	0.8998	0.8498	0.9892	0.8239	0.7611	0.8395	0.8561	0.9052
	Std	0.0036	0.0874	0.0929	0.0047	0.0954	0.0963	0.0723	0.0945	0.0881
Recall	Mean	0.9896	0.8782	0.8020	0.9888	0.7602	0.6780	0.8769	0.8166	0.8800
	Std	0.0038	0.0676	0.0723	0.0049	0.0556	0.0634	0.0423	0.0703	0.0699
F1 score	Mean	0.9896	0.8486	0.7598	0.9888	0.7140	0.6300	0.8399	0.7748	0.8506
	Std	0.0038	0.0876	0.0856	0.0049	0.0624	0.0652	0.0548	0.0844	0.0905
ARI	Mean	0.9765	0.8556	0.7482	0.9745	0.6763	0.5446	0.8574	0.7719	0.8563
	Std	0.0083	0.0661	0.1088	0.0111	0.0981	0.1018	0.0450	0.1005	0.0688

Table 6.1: $N = 2000$, $V = 6$, $L = 8$, $\alpha = 0.85$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.5947	0.5988	0.5896	0.5712	0.5791	0.5535	0.5055	0.5916	0.5970
	Std	0.0224	0.0216	0.0247	0.0357	0.0321	0.0316	0.0251	0.0218	0.0212
Precision	Mean	0.5402	0.5420	0.5345	0.5063	0.5258	0.4918	0.4137	0.5386	0.5405
	Std	0.0252	0.0205	0.0329	0.0571	0.0398	0.0475	0.0567	0.0272	0.0192
Recall	Mean	0.5947	0.5988	0.5896	0.5712	0.5791	0.5535	0.5055	0.5916	0.5970
	Std	0.0224	0.0216	0.0247	0.0357	0.0321	0.0316	0.0251	0.0218	0.0212
F1 score	Mean	0.5530	0.5571	0.5444	0.5145	0.5330	0.4958	0.4099	0.5474	0.5550
	Std	0.0235	0.0226	0.0299	0.0521	0.0361	0.0424	0.0344	0.0252	0.0221
ARI	Mean	0.3678	0.3718	0.3640	0.3701	0.3535	0.3283	0.3697	0.3655	0.3706
	Std	0.0229	0.0266	0.0270	0.0240	0.0340	0.0328	0.0232	0.0255	0.0258

Table 6.2: $N = 2000$, $V = 6$, $L = 8$, $\alpha = 0.85$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.6434	0.6496	0.6495	0.6380	0.6453	0.6303	0.5874	0.6512	0.6510
	Std	0.0191	0.0199	0.0196	0.0309	0.0219	0.0262	0.0294	0.0191	0.0196
Precision	Mean	0.6599	0.6600	0.6619	0.6444	0.6574	0.6458	0.5783	0.6626	0.6606
	Std	0.0209	0.0218	0.0223	0.0394	0.0209	0.0255	0.0551	0.0222	0.0202
Recall	Mean	0.6434	0.6496	0.6495	0.6380	0.6453	0.6303	0.5874	0.6512	0.6510
	Std	0.0191	0.0199	0.0196	0.0309	0.0219	0.0262	0.0294	0.0191	0.0196
F1 score	Mean	0.6388	0.6439	0.6430	0.6297	0.6399	0.6263	0.5548	0.6451	0.6446
	Std	0.0194	0.0206	0.0194	0.0384	0.0212	0.0269	0.0419	0.0201	0.0204
ARI	Mean	0.3846	0.3923	0.3926	0.3869	0.3879	0.3709	0.3856	0.3940	0.3945
	Std	0.0246	0.0260	0.0279	0.0260	0.0295	0.0330	0.0271	0.0259	0.0265

Table 7.1: $N = 2000$, $V = 15$, $L = 8$, $\alpha = 0.5$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9949	0.6652	0.6709	0.6745	0.6762	0.6612	0.6432	0.6713	0.6630
	Std	0.0037	0.0804	0.0874	0.0712	0.0874	0.1001	0.0704	0.0873	0.0786
Precision	Mean	0.9950	0.5585	0.5851	0.5540	0.5945	0.6018	0.5276	0.5821	0.5530
	Std	0.0036	0.0861	0.0833	0.0797	0.0811	0.0829	0.0833	0.0845	0.0851
Recall	Mean	0.9949	0.6652	0.6709	0.6745	0.6762	0.6612	0.6432	0.6713	0.6630
	Std	0.0037	0.0804	0.0874	0.0712	0.0874	0.1001	0.0704	0.0873	0.0786
F1 score	Mean	0.9949	0.5821	0.5988	0.5867	0.6068	0.5942	0.5575	0.5974	0.5781
	Std	0.0037	0.0911	0.0961	0.0813	0.0955	0.1076	0.0814	0.0964	0.0895
ARI	Mean	0.9885	0.5971	0.5678	0.6338	0.5719	0.5494	0.5572	0.5748	0.6011
	Std	0.0084	0.1275	0.1297	0.1158	0.1258	0.1364	0.1113	0.1309	0.1248

Table 7.2: $N = 2000$, $V = 15$, $L = 8$, $\alpha = 0.5$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9970	0.8502	0.8134	0.8852	0.8119	0.7981	0.8253	0.8216	0.8556
	Std	0.0030	0.0997	0.0963	0.0938	0.0931	0.0970	0.0907	0.0989	0.0988
Precision	Mean	0.9970	0.7971	0.7778	0.8347	0.7908	0.7905	0.7543	0.7897	0.8002
	Std	0.0029	0.1217	0.1169	0.1294	0.1205	0.1215	0.1255	0.1229	0.1222
Recall	Mean	0.9970	0.8502	0.8134	0.8852	0.8119	0.7981	0.8253	0.8216	0.8556
	Std	0.0030	0.0997	0.0963	0.0938	0.0931	0.0970	0.0907	0.0989	0.0988
F1 score	Mean	0.9970	0.8060	0.7668	0.8494	0.7660	0.7519	0.7757	0.7759	0.8116
	Std	0.0030	0.1253	0.1112	0.1221	0.1062	0.1083	0.1153	0.1171	0.1257
ARI	Mean	0.9931	0.8335	0.7662	0.8799	0.7640	0.7399	0.7923	0.7796	0.8439
	Std	0.0068	0.1158	0.1403	0.0975	0.1353	0.1436	0.1152	0.1396	0.1077

Table 8.1: $N = 2000$, $V = 15$, $L = 8$, $\alpha = 0.85$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.6737	0.6307	0.6105	0.6224	0.5946	0.5654	0.5919	0.6148	0.6399
	Std	0.0221	0.0307	0.0362	0.0423	0.0472	0.0534	0.0468	0.0397	0.0315
Precision	Mean	0.6241	0.5987	0.5828	0.5591	0.5701	0.5566	0.5462	0.5872	0.6147
	Std	0.0252	0.0377	0.0452	0.0703	0.0636	0.0694	0.0979	0.0490	0.0428
Recall	Mean	0.6737	0.6307	0.6105	0.6224	0.5946	0.5654	0.5919	0.6148	0.6399
	Std	0.0221	0.0307	0.0362	0.0423	0.0472	0.0534	0.0468	0.0397	0.0315
F1 score	Mean	0.6395	0.5871	0.5617	0.5686	0.5437	0.5194	0.5148	0.5659	0.5971
	Std	0.0235	0.0359	0.0403	0.0572	0.0564	0.0645	0.0704	0.0454	0.0411
ARI	Mean	0.4653	0.4296	0.4119	0.4278	0.3920	0.3497	0.4266	0.4147	0.4317
	Std	0.0316	0.0347	0.0498	0.0404	0.0595	0.0632	0.0285	0.0521	0.0350

Table 8.2: $N = 2000$, $V = 15$, $L = 8$, $\alpha = 0.85$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.7667	0.7662	0.7462	0.7284	0.7306	0.7039	0.6804	0.7598	0.7682
	Std	0.0176	0.0198	0.0369	0.0368	0.0398	0.0503	0.0300	0.0404	0.0208
Precision	Mean	0.7861	0.7860	0.7709	0.7508	0.7569	0.7355	0.7126	0.7813	0.7848
	Std	0.0174	0.0186	0.0280	0.0424	0.0363	0.0529	0.0442	0.0324	0.0195
Recall	Mean	0.7667	0.7662	0.7462	0.7284	0.7306	0.7039	0.6804	0.7598	0.7682
	Std	0.0176	0.0198	0.0369	0.0368	0.0398	0.0503	0.0300	0.0404	0.0208
F1 score	Mean	0.7686	0.7680	0.7481	0.7205	0.7316	0.7015	0.6536	0.7607	0.7691
	Std	0.0175	0.0193	0.0381	0.0472	0.0448	0.0629	0.0366	0.0420	0.0203
ARI	Mean	0.5415	0.5399	0.5150	0.5065	0.4977	0.4616	0.4968	0.5352	0.5452
	Std	0.0295	0.0340	0.0487	0.0409	0.0515	0.0553	0.0331	0.0547	0.0361

Table 9.1: $N = 2000$, $V = 6$, $L = 15$, $\alpha = 0.5$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.5115	0.4570	0.4700	0.4863	0.4624	0.4410	0.5077	0.4694	0.4602
	Std	0.0200	0.0305	0.0305	0.0383	0.0446	0.0559	0.0233	0.0309	0.0350
Precision	Mean	0.3076	0.2902	0.3269	0.3022	0.3374	0.3429	0.3491	0.3254	0.2913
	Std	0.0208	0.0355	0.0440	0.0440	0.0448	0.0428	0.0244	0.0421	0.0381
Recall	Mean	0.5115	0.4570	0.4700	0.4863	0.4624	0.4410	0.5077	0.4694	0.4602
	Std	0.0200	0.0305	0.0305	0.0383	0.0446	0.0559	0.0233	0.0309	0.0350
F1 score	Mean	0.3705	0.3322	0.3575	0.3546	0.3555	0.3401	0.3908	0.3578	0.3328
	Std	0.0215	0.0312	0.0393	0.0437	0.0492	0.0542	0.0246	0.0389	0.0370
ARI	Mean	0.5668	0.4315	0.4075	0.4896	0.3801	0.3386	0.4900	0.4038	0.4434
	Std	0.0312	0.0388	0.0345	0.0311	0.0591	0.0799	0.0236	0.0374	0.0389

Table 9.2: $N = 2000$, $V = 6$, $L = 15$, $\alpha = 0.5$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9004	0.7271	0.6941	0.8026	0.6805	0.6535	0.7158	0.6920	0.7435
	Std	0.0291	0.0601	0.0388	0.0443	0.0532	0.0675	0.0603	0.0413	0.0585
Precision	Mean	0.8595	0.6603	0.6531	0.7310	0.6545	0.6648	0.6242	0.6537	0.6718
	Std	0.0413	0.0730	0.0614	0.0698	0.0665	0.0666	0.0744	0.0640	0.0644
Recall	Mean	0.9004	0.7271	0.6941	0.8026	0.6805	0.6535	0.7158	0.6920	0.7435
	Std	0.0291	0.0601	0.0388	0.0443	0.0532	0.0675	0.0603	0.0413	0.0585
F1 score	Mean	0.8731	0.6605	0.6272	0.7479	0.6155	0.5935	0.6361	0.6221	0.6785
	Std	0.0381	0.0698	0.0460	0.0572	0.0557	0.0686	0.0782	0.0487	0.0667
ARI	Mean	0.8580	0.6716	0.6239	0.7599	0.5892	0.5153	0.7082	0.6270	0.6903
	Std	0.0372	0.0648	0.0518	0.0477	0.0831	0.1131	0.0466	0.0532	0.0672

Table 10.1: $N = 2000$, $V = 6$, $L = 15$, $\alpha = 0.85$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.3444	0.3508	0.3534	0.3390	0.3477	0.3379	0.3381	0.3527	0.3489
	Std	0.0193	0.0187	0.0179	0.0200	0.0185	0.0194	0.0204	0.0180	0.0188
Precision	Mean	0.1858	0.2074	0.2107	0.1812	0.2083	0.2070	0.1776	0.2098	0.2036
	Std	0.0214	0.0178	0.0174	0.0243	0.0179	0.0187	0.0213	0.0167	0.0196
Recall	Mean	0.3444	0.3508	0.3534	0.3390	0.3477	0.3379	0.3381	0.3527	0.3489
	Std	0.0193	0.0187	0.0179	0.0200	0.0185	0.0194	0.0204	0.0180	0.0188
F1 score	Mean	0.2328	0.2464	0.2489	0.2260	0.2448	0.2385	0.2223	0.2485	0.2429
	Std	0.0193	0.0178	0.0170	0.0212	0.0177	0.0177	0.0191	0.0173	0.0193
ARI	Mean	0.2331	0.2371	0.2372	0.2301	0.2312	0.2116	0.2339	0.2383	0.2387
	Std	0.0179	0.0171	0.0167	0.0184	0.0168	0.0227	0.0196	0.0176	0.0173

Table 10.2: $N = 2000$, $V = 6$, $L = 15$, $\alpha = 0.85$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.4993	0.5151	0.5116	0.4835	0.5052	0.4783	0.4888	0.5140	0.5153
	Std	0.0206	0.0196	0.0190	0.0258	0.0211	0.0335	0.0295	0.0210	0.0198
Precision	Mean	0.4565	0.4842	0.4910	0.4274	0.4882	0.4671	0.4284	0.4928	0.4825
	Std	0.0267	0.0245	0.0233	0.0436	0.0249	0.0330	0.0466	0.0241	0.0260
Recall	Mean	0.4993	0.5151	0.5116	0.4835	0.5052	0.4783	0.4888	0.5140	0.5153
	Std	0.0206	0.0196	0.0190	0.0258	0.0211	0.0335	0.0295	0.0210	0.0198
F1 score	Mean	0.4615	0.4812	0.4803	0.4330	0.4740	0.4466	0.4369	0.4825	0.4801
	Std	0.0225	0.0203	0.0184	0.0335	0.0200	0.0335	0.0377	0.0206	0.0220
ARI	Mean	0.2926	0.2918	0.2852	0.2805	0.2804	0.2632	0.2867	0.2868	0.2946
	Std	0.0200	0.0212	0.0208	0.0230	0.0218	0.0272	0.0255	0.0224	0.0204

Table 11.1: $N = 2000$, $V = 15$, $L = 15$, $\alpha = 0.5$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.5082	0.3900	0.4049	0.3906	0.3807	0.3215	0.3948	0.4012	0.3942
	Std	0.0171	0.0434	0.0567	0.0191	0.0708	0.0678	0.0193	0.0592	0.0438
Precision	Mean	0.2977	0.2950	0.3126	0.2933	0.3083	0.3096	0.3018	0.3086	0.2944
	Std	0.0186	0.0406	0.0537	0.0233	0.0580	0.0512	0.0278	0.0509	0.0394
Recall	Mean	0.5082	0.3900	0.4049	0.3906	0.3807	0.3215	0.3948	0.4012	0.3942
	Std	0.0171	0.0434	0.0567	0.0191	0.0708	0.0678	0.0193	0.0592	0.0438
F1 score	Mean	0.3651	0.3045	0.3227	0.3066	0.2989	0.2570	0.3117	0.3168	0.3085
	Std	0.0189	0.0385	0.0549	0.0182	0.0698	0.0597	0.0213	0.0571	0.0380
ARI	Mean	0.5639	0.2347	0.2426	0.1957	0.2269	0.1505	0.2166	0.2471	0.2452
	Std	0.0135	0.0685	0.0792	0.0289	0.0818	0.0775	0.0481	0.0817	0.0738

Table 11.2: $N = 2000$, $V = 15$, $L = 15$, $\alpha = 0.5$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9023	0.5426	0.5552	0.5296	0.5362	0.4595	0.5228	0.5557	0.5494
	Std	0.0142	0.0667	0.0793	0.0321	0.0843	0.0891	0.0290	0.0797	0.0728
Precision	Mean	0.8728	0.4791	0.5090	0.4541	0.5087	0.4975	0.4379	0.5129	0.4852
	Std	0.0259	0.0699	0.0748	0.0415	0.0761	0.0653	0.0425	0.0673	0.0692
Recall	Mean	0.9023	0.5426	0.5552	0.5296	0.5362	0.4595	0.5228	0.5557	0.5494
	Std	0.0142	0.0667	0.0793	0.0321	0.0843	0.0891	0.0290	0.0797	0.0728
F1 score	Mean	0.8794	0.4712	0.4903	0.4605	0.4699	0.4023	0.4476	0.4895	0.4779
	Std	0.0190	0.0690	0.0804	0.0375	0.0882	0.0832	0.0376	0.0812	0.0736
ARI	Mean	0.8768	0.3928	0.4045	0.3082	0.3656	0.2599	0.3338	0.4040	0.4120
	Std	0.0200	0.0963	0.1129	0.0599	0.1117	0.1135	0.0732	0.1177	0.1016

Table 12.1: $N = 2000$, $V = 15$, $L = 15$, $\alpha = 0.85$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.3972	0.3825	0.3836	0.3289	0.3826	0.3801	0.3549	0.3817	0.3832
	Std	0.0250	0.0228	0.0197	0.0330	0.0202	0.0238	0.0250	0.0193	0.0227
Precision	Mean	0.2376	0.2710	0.2768	0.1873	0.2769	0.2816	0.2387	0.2744	0.2722
	Std	0.0204	0.0255	0.0268	0.0365	0.0249	0.0280	0.0234	0.0267	0.0253
Recall	Mean	0.3972	0.3825	0.3836	0.3289	0.3826	0.3801	0.3549	0.3817	0.3832
	Std	0.0250	0.0228	0.0197	0.0330	0.0202	0.0238	0.0250	0.0193	0.0227
F1 score	Mean	0.2883	0.2984	0.3010	0.2183	0.3001	0.2991	0.2575	0.2990	0.2991
	Std	0.0222	0.0213	0.0197	0.0361	0.0192	0.0229	0.0261	0.0195	0.0208
ARI	Mean	0.2612	0.2066	0.2035	0.2198	0.2024	0.1957	0.2281	0.2034	0.2077
	Std	0.0216	0.0223	0.0243	0.0238	0.0236	0.0207	0.0232	0.0233	0.0255

Table 12.2: $N = 2000$, $V = 15$, $L = 15$, $\alpha = 0.85$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.5573	0.5150	0.5042	0.4682	0.5032	0.4984	0.4730	0.5088	0.5171
	Std	0.0311	0.0254	0.0249	0.0479	0.0250	0.0268	0.0384	0.0262	0.0289
Precision	Mean	0.5278	0.5298	0.5132	0.4113	0.5157	0.5089	0.4343	0.5195	0.5364
	Std	0.0430	0.0327	0.0341	0.0737	0.0360	0.0368	0.0605	0.0369	0.0387
Recall	Mean	0.5573	0.5150	0.5042	0.4682	0.5032	0.4984	0.4730	0.5088	0.5171
	Std	0.0311	0.0254	0.0249	0.0479	0.0250	0.0268	0.0384	0.0262	0.0289
F1 score	Mean	0.5251	0.4900	0.4778	0.4030	0.4759	0.4704	0.4082	0.4795	0.4914
	Std	0.0363	0.0266	0.0260	0.0592	0.0255	0.0257	0.0507	0.0272	0.0290
ARI	Mean	0.3485	0.2821	0.2793	0.2980	0.2772	0.2735	0.3072	0.2819	0.2824
	Std	0.0263	0.0308	0.0304	0.0364	0.0279	0.0288	0.0318	0.0313	0.0346

Real-World UCI Data

Datasets from the UCI Machine Learning Repository used in the experiments.

No.	Dataset	N	V	L
1	Breast Cancer (Prognostic)	194	32	2
2	Balance Scale	625	4	3
3	Car Evaluation	1728	6	4
4	Connectionist Bench Sonar	208	60	2
5	Contraceptive Method Choice	1473	9	3
6	Hayes-Roth	132	4	3
7	Hepatitis	80	19	2
8	Ionosphere	351	34	2
9	Glass	214	9	6
10	Congressional Voting Records	232	16	2
11	Spambase	4601	57	2
12	Mushroom	8214	22	7
13	Yeast	1484	8	10
14	Bach Choral Harmony	5665	15	102
15	Led Display Domain	1000	7	10

Table 1.1. Breast Cancer (Prognostic), tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.7110	0.6955	0.6967	0.6955	0.6959	0.6967	0.6947	0.6959	0.6963
	Std	0.0818	0.0731	0.0726	0.0723	0.0720	0.0720	0.0736	0.0708	0.0725
Precision	Mean	0.6457	0.6558	0.6565	0.6561	0.6563	0.6572	0.6559	0.6562	0.6547
	Std	0.0780	0.0874	0.0887	0.0870	0.0869	0.0872	0.0874	0.0866	0.0873
Recall	Mean	0.7110	0.6955	0.6967	0.6955	0.6959	0.6967	0.6947	0.6959	0.6963
	Std	0.0818	0.0731	0.0726	0.0723	0.0720	0.0720	0.0736	0.0708	0.0725
F1 score	Mean	0.6657	0.6646	0.6659	0.6647	0.6650	0.6658	0.6644	0.6650	0.6648
	Std	0.0736	0.0783	0.0782	0.0778	0.0776	0.0779	0.0784	0.0770	0.0777
ARI	Mean	0.0051	0.0175	0.0207	0.0174	0.0176	0.0192	0.0177	0.0170	0.0177
	Std	0.0586	0.0887	0.0913	0.0888	0.0887	0.0869	0.0889	0.0883	0.0899

Table 1.2. Breast Cancer (Prognostic), tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.6967	0.6629	0.6633	0.6625	0.6620	0.6625	0.6600	0.6633	0.6608
	Std	0.0834	0.0639	0.0650	0.0657	0.0658	0.0652	0.0665	0.0634	0.0661
Precision	Mean	0.6636	0.6516	0.6528	0.6517	0.6516	0.6520	0.6505	0.6511	0.6508
	Std	0.0947	0.0800	0.0789	0.0780	0.0780	0.0779	0.0783	0.0784	0.0782
Recall	Mean	0.6967	0.6629	0.6633	0.6625	0.6620	0.6625	0.6600	0.6633	0.6608
	Std	0.0834	0.0639	0.0650	0.0657	0.0658	0.0652	0.0665	0.0634	0.0661
F1 score	Mean	0.6676	0.6499	0.6512	0.6500	0.6497	0.6502	0.6483	0.6500	0.6489
	Std	0.0843	0.0696	0.0699	0.0698	0.0698	0.0695	0.0704	0.0690	0.0701
ARI	Mean	0.0304	0.0079	0.0090	0.0073	0.0073	0.0077	0.0061	0.0065	0.0064
	Std	0.0881	0.0650	0.0614	0.0621	0.0621	0.0621	0.0652	0.0623	0.0617

Table 2.1. Balance Scale, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.6989	0.7092	0.7055	0.7085	0.7056	0.7046	0.7118	0.7108	0.7121
	Std	0.0266	0.0326	0.0319	0.0310	0.0314	0.0317	0.0304	0.0307	0.0351
Precision	Mean	0.6511	0.6612	0.6573	0.6600	0.6571	0.6574	0.6651	0.6639	0.6648
	Std	0.0316	0.0370	0.0353	0.0355	0.0342	0.0351	0.0337	0.0335	0.0364
Recall	Mean	0.6989	0.7092	0.7055	0.7085	0.7056	0.7046	0.7118	0.7108	0.7121
	Std	0.0266	0.0326	0.0319	0.0310	0.0314	0.0317	0.0304	0.0307	0.0351
F1 score	Mean	0.6707	0.6811	0.6774	0.6804	0.6774	0.6764	0.6835	0.6825	0.6839
	Std	0.0292	0.0344	0.0338	0.0326	0.0331	0.0339	0.0316	0.0316	0.0352
ARI	Mean	0.2219	0.2427	0.2361	0.2414	0.2364	0.2339	0.2473	0.2454	0.2494
	Std	0.0457	0.0606	0.0581	0.0568	0.0581	0.0574	0.0600	0.0608	0.0713

Table 2.2. Balance Scale, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.7442	0.7825	0.7767	0.7782	0.7753	0.7720	0.7761	0.7833	0.7847
	Std	0.0261	0.0320	0.0334	0.0310	0.0308	0.0315	0.0276	0.0255	0.0279
Precision	Mean	0.7067	0.7334	0.7331	0.7295	0.7302	0.7288	0.7277	0.7341	0.7341
	Std	0.0326	0.0357	0.0383	0.0328	0.0391	0.0395	0.0299	0.0316	0.0305
Recall	Mean	0.7442	0.7825	0.7767	0.7782	0.7753	0.7720	0.7761	0.7833	0.7847
	Std	0.0261	0.0320	0.0334	0.0310	0.0308	0.0315	0.0276	0.0255	0.0279
F1 score	Mean	0.7199	0.7551	0.7502	0.7508	0.7485	0.7454	0.7487	0.7558	0.7567
	Std	0.0280	0.0332	0.0342	0.0310	0.0324	0.0334	0.0277	0.0274	0.0283
ARI	Mean	0.3239	0.4173	0.4031	0.4063	0.3981	0.3900	0.4000	0.4176	0.4212
	Std	0.0633	0.0757	0.0773	0.0744	0.0726	0.0745	0.0672	0.0625	0.0697

Table 3.1. Car Evaluation, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.7887	0.7878	0.7863	0.7887	0.7896	0.7969	0.7887	0.7872	0.7887
	Std	0.0190	0.0190	0.0166	0.0190	0.0150	0.0147	0.0190	0.0181	0.0190
Precision	Mean	0.7300	0.7300	0.7330	0.7300	0.7452	0.7670	0.7300	0.7301	0.7300
	Std	0.0271	0.0271	0.0278	0.0271	0.0306	0.0207	0.0271	0.0274	0.0271
Recall	Mean	0.7887	0.7878	0.7863	0.7887	0.7896	0.7969	0.7887	0.7872	0.7887
	Std	0.0190	0.0190	0.0166	0.0190	0.0150	0.0147	0.0190	0.0181	0.0190
F1 score	Mean	0.7554	0.7549	0.7554	0.7554	0.7627	0.7759	0.7554	0.7545	0.7554
	Std	0.0172	0.0170	0.0164	0.0172	0.0197	0.0164	0.0172	0.0161	0.0172
ARI	Mean	0.5088	0.5061	0.4982	0.5088	0.5008	0.5109	0.5088	0.5041	0.5088
	Std	0.0527	0.0530	0.0480	0.0527	0.0389	0.0379	0.0527	0.0532	0.0527

Table 3.2. Car Evaluation, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.8339	0.8443	0.8422	0.8443	0.8363	0.8304	0.8443	0.8444	0.8443
	Std	0.0183	0.0151	0.0183	0.0151	0.0193	0.0182	0.0151	0.0152	0.0151
Precision	Mean	0.8523	0.8451	0.8434	0.8458	0.8385	0.8319	0.8458	0.8445	0.8458
	Std	0.0207	0.0223	0.0214	0.0222	0.0181	0.0144	0.0222	0.0224	0.0222
Recall	Mean	0.8339	0.8443	0.8422	0.8443	0.8363	0.8304	0.8443	0.8444	0.8443
	Std	0.0183	0.0151	0.0183	0.0151	0.0193	0.0182	0.0151	0.0152	0.0151
F1 score	Mean	0.8308	0.8371	0.8339	0.8376	0.8254	0.8160	0.8376	0.8367	0.8376
	Std	0.0198	0.0186	0.0219	0.0181	0.0224	0.0176	0.0181	0.0188	0.0181
ARI	Mean	0.6506	0.6821	0.6665	0.6836	0.6351	0.6047	0.6836	0.6794	0.6836
	Std	0.0554	0.0489	0.0671	0.0464	0.0720	0.0553	0.0464	0.0534	0.0464

Table 4.1. Connectionist Bench Sonar, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.7211	0.7162	0.7158	0.7165	0.7135	0.7138	0.7138	0.7158	0.7127
	Std	0.0674	0.0671	0.0659	0.0667	0.0660	0.0661	0.0661	0.0681	0.0651
Precision	Mean	0.7447	0.7292	0.7301	0.7300	0.7270	0.7275	0.7274	0.7294	0.7263
	Std	0.0725	0.0703	0.0678	0.0697	0.0681	0.0689	0.0690	0.0707	0.0679
Recall	Mean	0.7211	0.7162	0.7158	0.7165	0.7135	0.7138	0.7138	0.7158	0.7127
	Std	0.0674	0.0671	0.0659	0.0667	0.0660	0.0661	0.0661	0.0681	0.0651
F1 score	Mean	0.7166	0.7133	0.7125	0.7136	0.7106	0.7108	0.7108	0.7128	0.7097
	Std	0.0690	0.0688	0.0675	0.0685	0.0674	0.0679	0.0679	0.0698	0.0668
ARI	Mean	0.1978	0.1892	0.1879	0.1897	0.1839	0.1846	0.1846	0.1891	0.1821
	Std	0.1196	0.1148	0.1128	0.1141	0.1122	0.1125	0.1125	0.1172	0.1101

Table 4.2. Connectionist Bench Sonar, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.7235	0.7131	0.7135	0.7115	0.7119	0.7108	0.7131	0.7127	0.7119
	Std	0.0696	0.0655	0.0662	0.0658	0.0635	0.0659	0.0647	0.0644	0.0638
Precision	Mean	0.7366	0.7229	0.7239	0.7215	0.7216	0.7206	0.7222	0.7228	0.7214
	Std	0.0685	0.0655	0.0650	0.0663	0.0630	0.0657	0.0646	0.0641	0.0632
Recall	Mean	0.7235	0.7131	0.7135	0.7115	0.7119	0.7108	0.7131	0.7127	0.7119
	Std	0.0696	0.0655	0.0662	0.0658	0.0635	0.0659	0.0647	0.0644	0.0638
F1 score	Mean	0.7220	0.7118	0.7119	0.7103	0.7106	0.7095	0.7118	0.7114	0.7106
	Std	0.0697	0.0667	0.0676	0.0669	0.0648	0.0672	0.0659	0.0657	0.0652
ARI	Mean	0.2037	0.1830	0.1840	0.1804	0.1799	0.1792	0.1825	0.1817	0.1801
	Std	0.1232	0.1125	0.1173	0.1142	0.1106	0.1126	0.1112	0.1127	0.1110

Table 5.1. Contraceptive Method Choice, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.5016	0.5118	0.5128	0.5131	0.5133	0.5130	0.5215	0.5109	0.5143
	Std	0.0293	0.0269	0.0291	0.0279	0.0316	0.0296	0.0246	0.0257	0.0267
Precision	Mean	0.5461	0.5677	0.5374	0.5754	0.5147	0.5043	0.5677	0.5604	0.5764
	Std	0.0773	0.0564	0.0677	0.0524	0.0656	0.0734	0.0604	0.0581	0.0510
Recall	Mean	0.5016	0.5118	0.5128	0.5131	0.5133	0.5130	0.5215	0.5109	0.5143
	Std	0.0293	0.0269	0.0291	0.0279	0.0316	0.0296	0.0246	0.0257	0.0267
F1 score	Mean	0.4855	0.4992	0.4894	0.5052	0.4886	0.4873	0.5146	0.4942	0.5069
	Std	0.0428	0.0317	0.0389	0.0348	0.0428	0.0437	0.0387	0.0343	0.0345
ARI	Mean	0.0605	0.0676	0.0687	0.0698	0.0735	0.0732	0.0808	0.0659	0.0709
	Std	0.0247	0.0229	0.0284	0.0244	0.0279	0.0258	0.0223	0.0230	0.0241

Table 5.2. Contraceptive Method Choice, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.5431	0.5512	0.5433	0.5497	0.5373	0.5306	0.5492	0.5477	0.5524
	Std	0.0258	0.0237	0.0252	0.0227	0.0241	0.0268	0.0223	0.0246	0.0209
Precision	Mean	0.5619	0.5648	0.5594	0.5618	0.5600	0.5565	0.5652	0.5625	0.5628
	Std	0.0367	0.0284	0.0291	0.0260	0.0295	0.0401	0.0263	0.0291	0.0262
Recall	Mean	0.5431	0.5512	0.5433	0.5497	0.5373	0.5306	0.5492	0.5477	0.5524
	Std	0.0258	0.0237	0.0252	0.0227	0.0241	0.0268	0.0223	0.0246	0.0209
F1 score	Mean	0.5370	0.5455	0.5363	0.5450	0.5306	0.5220	0.5469	0.5419	0.5472
	Std	0.0322	0.0244	0.0258	0.0242	0.0238	0.0305	0.0232	0.0250	0.0232
ARI	Mean	0.1020	0.1115	0.1035	0.1111	0.0953	0.0886	0.1115	0.1084	0.1141
	Std	0.0298	0.0266	0.0294	0.0260	0.0262	0.0266	0.0259	0.0281	0.0244

Table 6.1. Hayes-Roth, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.5528	0.5503	0.5370	0.5528	0.5364	0.5479	0.5528	0.5358	0.5528
	Std	0.0602	0.0628	0.0437	0.0602	0.0536	0.0635	0.0602	0.0443	0.0602
Precision	Mean	0.3704	0.3730	0.5486	0.3704	0.5613	0.5848	0.3704	0.5169	0.3704
	Std	0.0585	0.0605	0.1309	0.0585	0.1251	0.1242	0.0585	0.1364	0.0585
Recall	Mean	0.5528	0.5503	0.5370	0.5528	0.5364	0.5479	0.5528	0.5358	0.5528
	Std	0.0602	0.0628	0.0437	0.0602	0.0536	0.0635	0.0602	0.0443	0.0602
F1 score	Mean	0.4241	0.4243	0.5131	0.4241	0.5189	0.5337	0.4241	0.4952	0.4241
	Std	0.0630	0.0630	0.0843	0.0630	0.0886	0.0895	0.0630	0.0843	0.0630
ARI	Mean	0.4657	0.4584	0.1953	0.4657	0.1766	0.1751	0.4657	0.2334	0.4657
	Std	0.0698	0.0893	0.1582	0.0698	0.1362	0.1309	0.0698	0.1777	0.0698

Table 6.2. Hayes-Roth, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.6564	0.6564	0.5952	0.6564	0.6194	0.6303	0.6564	0.6000	0.6564
	Std	0.0448	0.0448	0.0907	0.0448	0.0964	0.0913	0.0448	0.0831	0.0448
Precision	Mean	0.6759	0.6759	0.6597	0.6759	0.7019	0.7028	0.6759	0.6529	0.6759
	Std	0.0625	0.0625	0.0830	0.0625	0.0920	0.0964	0.0625	0.0763	0.0625
Recall	Mean	0.6564	0.6564	0.5952	0.6564	0.6194	0.6303	0.6564	0.6000	0.6564
	Std	0.0448	0.0448	0.0907	0.0448	0.0964	0.0913	0.0448	0.0831	0.0448
F1 score	Mean	0.6551	0.6551	0.5896	0.6551	0.6095	0.6200	0.6551	0.5979	0.6551
	Std	0.0476	0.0476	0.0942	0.0476	0.0967	0.0941	0.0476	0.0872	0.0476
ARI	Mean	0.3449	0.3449	0.2539	0.3449	0.2550	0.2709	0.3449	0.2583	0.3449
	Std	0.0757	0.0757	0.1180	0.0757	0.1244	0.1087	0.0757	0.1053	0.0757

Table 7.1. Hepatitis, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.8150	0.8110	0.8100	0.8080	0.8090	0.8100	0.8130	0.8170	0.8090
	Std	0.0743	0.0666	0.0700	0.0681	0.0676	0.0693	0.0699	0.0704	0.0683
Precision	Mean	0.8427	0.8166	0.8134	0.8137	0.8152	0.8138	0.8215	0.8245	0.8138
	Std	0.0893	0.0977	0.1014	0.1001	0.0991	0.1002	0.0988	0.0973	0.1002
Recall	Mean	0.8150	0.8110	0.8100	0.8080	0.8090	0.8100	0.8130	0.8170	0.8090
	Std	0.0743	0.0666	0.0700	0.0681	0.0676	0.0693	0.0699	0.0704	0.0683
F1 score	Mean	0.8155	0.8071	0.8050	0.8038	0.8053	0.8049	0.8098	0.8136	0.8044
	Std	0.0809	0.0783	0.0822	0.0805	0.0794	0.0814	0.0808	0.0798	0.0809
ARI	Mean	0.2458	0.1992	0.1947	0.1937	0.1970	0.1941	0.2141	0.2256	0.1938
	Std	0.2302	0.1894	0.1916	0.1905	0.1902	0.1901	0.1930	0.1989	0.1905

Table 7.2. Hepatitis, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.8120	0.8100	0.8080	0.8060	0.8080	0.8090	0.8060	0.8090	0.8080
	Std	0.0637	0.0686	0.0717	0.0712	0.0703	0.0705	0.0712	0.0705	0.0703
Precision	Mean	0.8377	0.8280	0.8242	0.8233	0.8267	0.8252	0.8257	0.8283	0.8252
	Std	0.0864	0.0919	0.0964	0.0961	0.0935	0.0949	0.0961	0.0941	0.0948
Recall	Mean	0.8120	0.8100	0.8080	0.8060	0.8080	0.8090	0.8060	0.8090	0.8080
	Std	0.0637	0.0686	0.0717	0.0712	0.0703	0.0705	0.0712	0.0705	0.0703
F1 score	Mean	0.8131	0.8129	0.8101	0.8080	0.8109	0.8108	0.8088	0.8118	0.8101
	Std	0.0746	0.0763	0.0807	0.0797	0.0777	0.0797	0.0794	0.0771	0.0791
ARI	Mean	0.2218	0.2294	0.2242	0.2178	0.2260	0.2257	0.2221	0.2296	0.2240
	Std	0.2077	0.1911	0.1944	0.1987	0.1926	0.1931	0.1943	0.1920	0.1931

Table 8.1. Ionosphere, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.8973	0.8866	0.8870	0.8868	0.8866	0.8866	0.8859	0.8866	0.8866
	Std	0.0382	0.0418	0.0408	0.0415	0.0418	0.0418	0.0421	0.0418	0.0418
Precision	Mean	0.9018	0.8924	0.8931	0.8927	0.8924	0.8924	0.8917	0.8924	0.8924
	Std	0.0332	0.0368	0.0351	0.0362	0.0368	0.0368	0.0373	0.0368	0.0368
Recall	Mean	0.8973	0.8866	0.8870	0.8868	0.8866	0.8866	0.8859	0.8866	0.8866
	Std	0.0382	0.0418	0.0408	0.0415	0.0418	0.0418	0.0421	0.0418	0.0418
F1 score	Mean	0.8967	0.8857	0.8861	0.8859	0.8857	0.8857	0.8850	0.8857	0.8857
	Std	0.0405	0.0445	0.0435	0.0442	0.0445	0.0445	0.0448	0.0445	0.0445
ARI	Mean	0.6293	0.5963	0.5974	0.5970	0.5963	0.5963	0.5942	0.5963	0.5963
	Std	0.1167	0.1265	0.1242	0.1255	0.1265	0.1265	0.1271	0.1265	0.1265

Table 8.2. Ionosphere, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.8789	0.8820	0.8816	0.8820	0.8816	0.8816	0.8811	0.8820	0.8823
	Std	0.0387	0.0322	0.0315	0.0318	0.0322	0.0322	0.0321	0.0321	0.0317
Precision	Mean	0.8846	0.8860	0.8857	0.8862	0.8854	0.8854	0.8851	0.8858	0.8861
	Std	0.0380	0.0323	0.0313	0.0316	0.0323	0.0323	0.0321	0.0324	0.0319
Recall	Mean	0.8789	0.8820	0.8816	0.8820	0.8816	0.8816	0.8811	0.8820	0.8823
	Std	0.0387	0.0322	0.0315	0.0318	0.0322	0.0322	0.0321	0.0321	0.0317
F1 score	Mean	0.8758	0.8795	0.8791	0.8795	0.8791	0.8791	0.8785	0.8796	0.8798
	Std	0.0396	0.0335	0.0327	0.0330	0.0334	0.0334	0.0333	0.0333	0.0330
ARI	Mean	0.5676	0.5757	0.5742	0.5756	0.5743	0.5743	0.5728	0.5757	0.5763
	Std	0.1178	0.1016	0.1002	0.0998	0.1016	0.1016	0.1013	0.1013	0.1003

Table 9.1. Glass, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.6289	0.6533	0.6437	0.5711	0.6148	0.6037	0.5607	0.6585	0.6596
	Std	0.0553	0.0512	0.0635	0.1033	0.0539	0.0477	0.1078	0.0581	0.0502
Precision	Mean	0.6315	0.6040	0.6239	0.5195	0.6043	0.5913	0.4859	0.6239	0.6000
	Std	0.0672	0.0646	0.0829	0.1792	0.0746	0.0704	0.1897	0.0799	0.0687
Recall	Mean	0.6289	0.6533	0.6437	0.5711	0.6148	0.6037	0.5607	0.6585	0.6596
	Std	0.0553	0.0512	0.0635	0.1033	0.0539	0.0477	0.1078	0.0581	0.0502
F1 score	Mean	0.6110	0.6134	0.6180	0.5028	0.5903	0.5771	0.4844	0.6251	0.6148
	Std	0.0577	0.0517	0.0677	0.1435	0.0589	0.0513	0.1505	0.0634	0.0547
ARI	Mean	0.2880	0.3154	0.3111	0.3092	0.2808	0.2613	0.3060	0.3273	0.3302
	Std	0.0744	0.0749	0.0863	0.0725	0.0723	0.0753	0.0773	0.0809	0.0830

Table 9.2. Glass, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.6589	0.6578	0.6544	0.6393	0.6315	0.6270	0.5907	0.6692	0.6652
	Std	0.0563	0.0569	0.0692	0.0763	0.0545	0.0567	0.1050	0.0576	0.0641
Precision	Mean	0.6763	0.6515	0.6649	0.6286	0.6455	0.6504	0.5599	0.6654	0.6459
	Std	0.0592	0.0790	0.0777	0.1033	0.0577	0.0552	0.1680	0.0732	0.0842
Recall	Mean	0.6589	0.6578	0.6544	0.6393	0.6315	0.6270	0.5907	0.6692	0.6652
	Std	0.0563	0.0569	0.0692	0.0763	0.0545	0.0567	0.1050	0.0576	0.0641
F1 score	Mean	0.6498	0.6386	0.6414	0.6138	0.6208	0.6174	0.5373	0.6529	0.6425
	Std	0.0587	0.0632	0.0736	0.0926	0.0551	0.0533	0.1437	0.0623	0.0698
ARI	Mean	0.3106	0.3119	0.3043	0.3047	0.2794	0.2718	0.3000	0.3308	0.3285
	Std	0.0732	0.0825	0.0954	0.0763	0.0727	0.0674	0.0838	0.0816	0.0830

Table 10.1. Congressional Voting Records, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9514	0.9493	0.9486	0.9493	0.9493	0.9493	0.9493	0.9493	0.9493
	Std	0.0269	0.0270	0.0266	0.0270	0.0270	0.0270	0.0270	0.0270	0.0270
Precision	Mean	0.9541	0.9524	0.9517	0.9524	0.9524	0.9524	0.9524	0.9524	0.9524
	Std	0.0241	0.0241	0.0237	0.0241	0.0241	0.0241	0.0241	0.0241	0.0241
Recall	Mean	0.9514	0.9493	0.9486	0.9493	0.9493	0.9493	0.9493	0.9493	0.9493
	Std	0.0269	0.0270	0.0266	0.0270	0.0270	0.0270	0.0270	0.0270	0.0270
F1 score	Mean	0.9513	0.9493	0.9486	0.9493	0.9493	0.9493	0.9493	0.9493	0.9493
	Std	0.0270	0.0271	0.0267	0.0271	0.0271	0.0271	0.0271	0.0271	0.0271
ARI	Mean	0.8146	0.8070	0.8044	0.8070	0.8070	0.8070	0.8070	0.8070	0.8070
	Std	0.0961	0.0965	0.0948	0.0965	0.0965	0.0965	0.0965	0.0965	0.0965

Table 10.2. Congressional Voting Records, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.9490	0.9483	0.9493	0.9493	0.9486	0.9486	0.9486	0.9483	0.9483
	Std	0.0269	0.0272	0.0277	0.0272	0.0275	0.0275	0.0275	0.0272	0.0272
Precision	Mean	0.9520	0.9513	0.9523	0.9521	0.9517	0.9517	0.9517	0.9513	0.9513
	Std	0.0231	0.0241	0.0247	0.0243	0.0244	0.0244	0.0244	0.0241	0.0241
Recall	Mean	0.9490	0.9483	0.9493	0.9493	0.9486	0.9486	0.9486	0.9483	0.9483
	Std	0.0269	0.0272	0.0277	0.0272	0.0275	0.0275	0.0275	0.0272	0.0272
F1 score	Mean	0.9490	0.9483	0.9493	0.9493	0.9486	0.9486	0.9486	0.9483	0.9483
	Std	0.0270	0.0272	0.0277	0.0273	0.0275	0.0275	0.0275	0.0272	0.0272
ARI	Mean	0.8058	0.8033	0.8072	0.8071	0.8046	0.8046	0.8046	0.8033	0.8033
	Std	0.0960	0.0969	0.0990	0.0975	0.0982	0.0982	0.0982	0.0969	0.0969

Table 11.1. Spambase, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.8693	0.8820	0.8820	0.8820	0.8820	0.8820	0.8820	0.8820	0.8820
	Std	0.0085	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111
Precision	Mean	0.8757	0.8832	0.8832	0.8832	0.8832	0.8832	0.8832	0.8832	0.8832
	Std	0.0084	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105
Recall	Mean	0.8693	0.8820	0.8820	0.8820	0.8820	0.8820	0.8820	0.8820	0.8820
	Std	0.0085	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111
F1 score	Mean	0.8667	0.8810	0.8810	0.8810	0.8810	0.8810	0.8810	0.8810	0.8810
	Std	0.0094	0.0113	0.0113	0.0113	0.0113	0.0113	0.0113	0.0113	0.0113
ARI	Mean	0.5430	0.5824	0.5824	0.5824	0.5824	0.5824	0.5824	0.5824	0.5824
	Std	0.0258	0.0337	0.0337	0.0337	0.0337	0.0337	0.0337	0.0337	0.0337

Table 11.2. Spambase, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.8982	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976
	Std	0.0086	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104
Precision	Mean	0.8997	0.8995	0.8995	0.8995	0.8995	0.8995	0.8995	0.8996	0.8996
	Std	0.0084	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105
Recall	Mean	0.8982	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976
	Std	0.0086	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104
F1 score	Mean	0.8969	0.8964	0.8964	0.8964	0.8964	0.8964	0.8964	0.8964	0.8964
	Std	0.0087	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104
ARI	Mean	0.6326	0.6310	0.6310	0.6309	0.6310	0.6310	0.6310	0.6310	0.6310
	Std	0.0272	0.0330	0.0330	0.0330	0.0330	0.0330	0.0330	0.0330	0.0330

Table 12.1. Mushroom, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.6069	0.6008	0.6012	0.5830	0.6181	0.6181	0.5807	0.6003	0.5970
	Std	0.0130	0.0112	0.0110	0.0133	0.0098	0.0098	0.0093	0.0110	0.0133
Precision	Mean	0.5392	0.5078	0.5063	0.5275	0.4701	0.4701	0.5154	0.5177	0.5334
	Std	0.0418	0.0635	0.0658	0.0490	0.0124	0.0124	0.0344	0.0690	0.0627
Recall	Mean	0.6069	0.6008	0.6012	0.5830	0.6181	0.6181	0.5807	0.6003	0.5970
	Std	0.0130	0.0112	0.0110	0.0133	0.0098	0.0098	0.0093	0.0110	0.0133
F1 score	Mean	0.5166	0.5262	0.5254	0.5106	0.5240	0.5240	0.5053	0.5292	0.5177
	Std	0.0213	0.0221	0.0229	0.0230	0.0117	0.0117	0.0237	0.0242	0.0245
ARI	Mean	0.2979	0.3096	0.3232	0.2755	0.3593	0.3593	0.2476	0.3076	0.2894
	Std	0.0198	0.0170	0.0198	0.0306	0.0120	0.0120	0.0277	0.0168	0.0206

Table 12.2. Mushroom, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.6403	0.6373	0.6362	0.6078	0.6377	0.6455	0.6111	0.6368	0.6352
	Std	0.0103	0.0123	0.0119	0.0171	0.0093	0.0101	0.0114	0.0124	0.0174
Precision	Mean	0.6432	0.5943	0.6030	0.6077	0.5951	0.5918	0.6431	0.5950	0.6332
	Std	0.0591	0.0415	0.0410	0.0501	0.0702	0.0694	0.0635	0.0395	0.0465
Recall	Mean	0.6403	0.6373	0.6362	0.6078	0.6377	0.6455	0.6111	0.6368	0.6352
	Std	0.0103	0.0123	0.0119	0.0171	0.0093	0.0101	0.0114	0.0124	0.0174
F1 score	Mean	0.5940	0.5995	0.6020	0.5418	0.5802	0.5846	0.5671	0.6000	0.5863
	Std	0.0298	0.0271	0.0269	0.0264	0.0268	0.0278	0.0310	0.0260	0.0340
ARI	Mean	0.3921	0.3995	0.3944	0.3254	0.3565	0.3712	0.3142	0.3991	0.3781
	Std	0.0290	0.0249	0.0257	0.0386	0.0120	0.0129	0.0251	0.0249	0.0311

Table 13.1. Yeast, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.5035	0.5598	0.5546	0.4481	0.5470	0.5397	0.4777	0.5591	0.5652
	Std	0.0268	0.0204	0.0200	0.0246	0.0185	0.0249	0.0329	0.0195	0.0219
Precision	Mean	0.3637	0.5488	0.5465	0.2620	0.5343	0.5271	0.3249	0.5494	0.5516
	Std	0.0440	0.0250	0.0235	0.0620	0.0204	0.0296	0.0568	0.0238	0.0242
Recall	Mean	0.5035	0.5598	0.5546	0.4481	0.5470	0.5397	0.4777	0.5591	0.5652
	Std	0.0268	0.0204	0.0200	0.0246	0.0185	0.0249	0.0329	0.0195	0.0219
F1 score	Mean	0.4093	0.5344	0.5295	0.3042	0.5243	0.5174	0.3627	0.5335	0.5401
	Std	0.0358	0.0232	0.0219	0.0262	0.0199	0.0266	0.0456	0.0226	0.0249
ARI	Mean	0.2700	0.2148	0.2106	0.2259	0.2043	0.2005	0.2557	0.2146	0.2199
	Std	0.0260	0.0214	0.0203	0.0217	0.0177	0.0195	0.0274	0.0204	0.0215

Table 13.2. Yeast, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.5690	0.5650	0.5632	0.4720	0.5590	0.5568	0.5099	0.5607	0.5633
	Std	0.0207	0.0204	0.0187	0.0292	0.0190	0.0231	0.0273	0.0187	0.0225
Precision	Mean	0.5631	0.5653	0.5626	0.3619	0.5569	0.5545	0.3883	0.5549	0.5582
	Std	0.0224	0.0247	0.0207	0.0709	0.0209	0.0260	0.0401	0.0230	0.0257
Recall	Mean	0.5690	0.5650	0.5632	0.4720	0.5590	0.5568	0.5099	0.5607	0.5633
	Std	0.0207	0.0204	0.0187	0.0292	0.0190	0.0231	0.0273	0.0187	0.0225
F1 score	Mean	0.5520	0.5461	0.5437	0.3485	0.5391	0.5364	0.4169	0.5401	0.5438
	Std	0.0231	0.0227	0.0214	0.0410	0.0215	0.0252	0.0281	0.0208	0.0258
ARI	Mean	0.2445	0.2365	0.2340	0.2424	0.2303	0.2280	0.2857	0.2285	0.2351
	Std	0.0224	0.0226	0.0203	0.0260	0.0197	0.0235	0.0242	0.0218	0.0225

Table 14.1. Bach Choral Harmony, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.4086	0.4162	0.4154	0.2711	0.4108	0.3967	0.3528	0.4154	0.4161
	Std	0.0233	0.0115	0.0145	0.0364	0.0176	0.0187	0.0200	0.0148	0.0109
Precision	Mean	0.2463	0.2659	0.2694	0.1152	0.2655	0.2493	0.1823	0.2705	0.2650
	Std	0.0211	0.0132	0.0166	0.0272	0.0200	0.0237	0.0229	0.0185	0.0140
Recall	Mean	0.4086	0.4162	0.4154	0.2711	0.4108	0.3967	0.3528	0.4154	0.4161
	Std	0.0233	0.0115	0.0145	0.0364	0.0176	0.0187	0.0200	0.0148	0.0109
F1 score	Mean	0.2930	0.3098	0.3123	0.1486	0.3078	0.2917	0.2302	0.3130	0.3091
	Std	0.0236	0.0122	0.0155	0.0361	0.0191	0.0218	0.0217	0.0170	0.0121
ARI	Mean	0.2690	0.2675	0.2545	0.1327	0.2532	0.2518	0.2083	0.2516	0.2678
	Std	0.0246	0.0163	0.0164	0.0306	0.0179	0.0178	0.0194	0.0161	0.0186

Table 14.2. Bach Choral Harmony, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.5416	0.5422	0.5267	0.3752	0.5210	0.5208	0.4617	0.5256	0.5456
	Std	0.0186	0.0121	0.0175	0.0574	0.0184	0.0187	0.0221	0.0167	0.0131
Precision	Mean	0.4174	0.4388	0.4273	0.2226	0.4211	0.4144	0.3095	0.4288	0.4359
	Std	0.0261	0.0186	0.0319	0.0624	0.0287	0.0271	0.0303	0.0355	0.0190
Recall	Mean	0.5416	0.5422	0.5267	0.3752	0.5210	0.5208	0.4617	0.5256	0.5456
	Std	0.0186	0.0121	0.0175	0.0574	0.0184	0.0187	0.0221	0.0167	0.0131
F1 score	Mean	0.4586	0.4687	0.4553	0.2608	0.4489	0.4455	0.3572	0.4556	0.4693
	Std	0.0239	0.0142	0.0233	0.0643	0.0225	0.0210	0.0269	0.0243	0.0149
ARI	Mean	0.3979	0.4051	0.3774	0.2183	0.3643	0.3689	0.3058	0.3717	0.4025
	Std	0.0252	0.0181	0.0282	0.0566	0.0379	0.0316	0.0277	0.0300	0.0243

Table 15.1. LED Display Domain, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.5616	0.5324	0.5152	0.5478	0.4764	0.4384	0.5418	0.5109	0.5379
	Std	0.0399	0.0357	0.0467	0.0468	0.0624	0.0579	0.0533	0.0568	0.0398
Precision	Mean	0.4861	0.4543	0.4316	0.4717	0.3888	0.3519	0.4618	0.4275	0.4629
	Std	0.0489	0.0518	0.0606	0.0611	0.0745	0.0706	0.0718	0.0749	0.0536
Recall	Mean	0.5616	0.5324	0.5152	0.5478	0.4764	0.4384	0.5418	0.5109	0.5379
	Std	0.0399	0.0357	0.0467	0.0468	0.0624	0.0579	0.0533	0.0568	0.0398
F1 score	Mean	0.5082	0.4749	0.4518	0.4928	0.4068	0.3635	0.4849	0.4477	0.4821
	Std	0.0465	0.0453	0.0552	0.0556	0.0733	0.0697	0.0649	0.0690	0.0481
ARI	Mean	0.3782	0.3494	0.3456	0.3609	0.3130	0.2767	0.3578	0.3380	0.3527
	Std	0.0403	0.0353	0.0448	0.0471	0.0554	0.0434	0.0532	0.0491	0.0385

Table 15.2. LED Display Domain, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l \log(p_l)$	$b = -p_l^{0.5} \log(p_l)$
Accuracy	Mean	0.6999	0.6735	0.6673	0.6711	0.6330	0.5946	0.6714	0.6602	0.6735
	Std	0.0290	0.0404	0.0518	0.0452	0.0624	0.0580	0.0475	0.0525	0.0389
Precision	Mean	0.7247	0.6937	0.6860	0.6920	0.6448	0.5998	0.6870	0.6793	0.6930
	Std	0.0361	0.0510	0.0672	0.0551	0.0873	0.0929	0.0611	0.0680	0.0506
Recall	Mean	0.6999	0.6735	0.6673	0.6711	0.6330	0.5946	0.6714	0.6602	0.6735
	Std	0.0290	0.0404	0.0518	0.0452	0.0624	0.0580	0.0475	0.0525	0.0389
F1 score	Mean	0.7022	0.6708	0.6625	0.6680	0.6158	0.5626	0.6663	0.6531	0.6704
	Std	0.0323	0.0473	0.0647	0.0519	0.0823	0.0770	0.0564	0.0666	0.0460
ARI	Mean	0.4485	0.4299	0.4271	0.4287	0.4076	0.3894	0.4332	0.4218	0.4307
	Std	0.0379	0.0430	0.0485	0.0454	0.0479	0.0464	0.0446	0.0463	0.0408